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Implicating with commitment

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Building on Geurts (2019a; b) programmatic proposal, this paper develops a commitment-based model of Scalar Implicatures (SIs) that shifts the explanatory focus from speaker beliefs and mental states to the dynamics between private commitments and their social expression. The model accounts for both standard scalar inferences and cases in which speakers' private stances remain partially opaque, offering a socially grounded alternative to belief-based accounts, with broad implications for the analysis of SIs across speech acts and the explanation of cross-cultural variation.

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1 Introduction

The mainstream view in the study of linguistic communication—what we might call the mentalist perspective—holds that communication is fundamentally a matter of exchanging information about the speakers’ communicative intentions (Grice 1957; 1975; Strawson 1964; Bach & Harnish 1979; Soames 1982; Sperber & Wilson 1986). On this view, to understand an utterance like “I’ll walk the dog”, the hearer must infer that the speaker intends to walk the dog and, crucially, intends for the hearer to recognise this intention. But this is not the only way to approach the purpose of such an utterance. Alternatively, it can be seen as expressing a commitment: in uttering “I’ll walk the dog”, the speaker enters into a normative relation with the hearer, committing themselves to walking the dog. This normative perspective, in effect, shifts the focus from a psychological model of communication to a social one—from seeing communication as the transmission of intentions and beliefs to seeing it as the sharing and negotiation of commitments (Brandom 1994; Kibble 2006a; b; Krifka 2015; Geurts 2019a; Krifka et al. 2023).

Needless to say, these two perspectives are compatible and many scholars, in fact, have emphasised the conceptual nexus between an individual’s doxastic and conative mental states and the kinds of commitments they are disposed to undertake (Seegerberg 1984; Bratman 1987; Brandom 1994; Clark 1996). Yet the normative perspective shifts the explanatory focus: it holds that the primary function of face-to-face communication is to establish, revise, and act upon shared commitments to facilitate the coordination of actions. This shift in perspective has been argued to address certain explanatory gaps found in intention-based models regarding both the evolution and acquisition of communicative competence (Geurts 2019a). Moreover, while intention-based theories often struggle to account for conventional speech acts—sometimes dismissing them as non-communicative (Bach & Harnish 1979)—a commitment-based approach can easily incorporate both conventional and non-conventional speech acts within a unified framework (Geurts 2019a).

Beyond speech acts, commitment-based models invite a broader re-examination of core pragmatic notions traditionally anchored in Gricean intentionalism. One such notion, for instance, is that of common ground. Traditionally thought of as a repository of mutual beliefs or knowledge (Stalnaker 1973; 1978; 2002; 2014), Geurts recently proposed to re-conceptualise the common ground as a network of mutual commitments (Geurts 2019a; 2020; 2024). Crucially, while mutual commitments have a recursive structure similar to that of mutual beliefs, they add a distinct normative dimension: rather than merely supporting interactions between interlocutors, as shared information does, mutual commitments also constrain and regulate their actions, enabling them to coordinate them. This raises a natural question: might other core notions in pragmatics be reinterpreted within a normative, commitment-based model?

One promising case is that of Scalar Implicatures (SIs). SIs occur when a speaker’s use of a less informative expression is interpreted as signalling that a more informative alternative is

unwarranted. For example, the utterance *The Dean met some of the students* may implicate that the Dean did not meet all of them, although the literal meaning of the statement does not exclude that possibility. Theories of SIs grounded in Grice’s influential framework typically treat such inferences as the product of speaker-oriented reasoning, based on assumptions about what the speaker believes and intends for the hearer to conclude. In these theories, the derivation of SIs crucially depends on attributing specific beliefs and intentions to the speaker. Geurts (2019a) proposes a commitment-based reinterpretation of the Gricean maxims, offering a proof of concept for the maxim of Quality and outlining a general schema for deriving implicatures within this framework. His proposal, however, remains underdeveloped, particularly with respect to the application of the maxim of Quantity, as well as other auxiliary assumptions central to SIs.

This paper develops this line of inquiry by investigating whether SIs—typically seen as inferences about what a speaker intends to convey or omit—can instead be analysed in terms of the commitments a speaker undertakes or withholds. Section 2 reviews the Gricean approach to Quantity implicatures, belief-based models of SIs, and Geurts’ programmatic proposal for a commitment-based alternative. Section 3 advances this alternative while identifying challenges concerning the strength and articulation of private and social commitments in scalar inferencing. Section 4 presents a revised account that focuses on the agent’s private commitments and their practical disposition to share these commitments with others. This account is shown to address the challenges identified in Section 3, and, in particular, to be well-suited to handle cases that Neo-Gricean models struggle to accommodate—cases where private commitments may be *partially* withheld to comply with other normative pressures (Fox 2014; Marty 2018; Agyemang 2020; Marty et al. 2022). Section 5 concludes by reflecting on the implications and potential extensions of the proposed account.

2 Conversational implicatures

2.1 Gricean foundations

Grice’s theory of conversational implicature provides a foundational framework for understanding how pragmatic inferences arise.¹ At the heart of this framework lies the Cooperative Principle, which posits that speakers generally aim to contribute to a conversation in ways that advance the shared goals of the conversational exchange. The underlying assumption is that participants in a conversation are engaged in a cooperative enterprise and that their utterances can be interpreted against this background of mutual purpose.

As is well known, Grice (1975) elaborates the Cooperative Principle through a set of four conversational maxims—Quality, Quantity, Relation, and Manner—each of which serves to

¹ We refer to Levinson (2000) and Horn (2004) for a comprehensive overview and interpretation of Grice’s framework and its place in pragmatic theory.

clarify the guiding principles that interlocutors are presumed to follow when producing and interpreting utterances. In this framework, conversational implicatures are those inferences licensed by the Cooperative Principle through the application of these maxims. Specifically, as Grice put it, an implicature is “what has to be supposed in order to preserve the supposition that the Cooperative Principle is being observed”. As Geurts (2019a) notes, while the maxims themselves are largely formulated in non-psychological terms—with the exception of Quality—the inferential mechanism by which implicatures are derived in Grice’s account relies heavily on nested attributions of belief and intention between interlocutors (e.g., Soames’s 1982 Gricean analysis of Quantity implicatures). To illustrate, consider the following utterance:

(1) I saw some animal in the woods.

Assuming the speaker is observing the maxims, the hearer may infer that the speaker intends to convey that they did see some animal, and therefore holds the corresponding belief (Quality). In addition, the speaker’s choice to leave which animal was seen unspecified invites further inference. Specifically, given Quantity, the fact that speaker could have, but did not utter a more specific statement (e.g., I saw *a deer*, I saw *a bear*) suggests that the speaker lacks adequate evidence to make such a claim. Thus, the hearer may infer that the speaker is uncertain about the identity of the animal and that this uncertainty is deliberately being communicated by leaving the description somewhat vague. More generally, Grice’s schema for implicature derivation can be summarised as follows:

1. The speaker uttered *p*.
2. Assuming they are observing the maxims, they would not have uttered *p* unless they believed *q*.
3. They have done nothing to prevent the hearer from inferring *q*.
4. Therefore, the speaker intends the hearer to infer that *q*, and thus implicates that *q*.

On this schema, the derivation of conversational implicatures is deeply tied to the interpretative process by which hearers recover speakers’ beliefs and communicative intentions. In this sense, intention recognition and mental-state attribution are not ancillary notions, but part of the explanatory core of how meaning is generated and recovered.

2.2 Belief-based accounts

The psychological framing of Grice’s theory has become standard in formal pragmatics, particularly in the treatment of SIs. For simplicity, we will refer to models that adopt Grice’s inferential schema and mentalistic vocabulary as *belief-based models*. Despite variations, these models share a core idea: implicature derivation hinges on reasoning about speakers’ mental states. To fix ideas, I will focus on the following formulations of the maxims of Quality and

Quantity, which I take to offer a faithful characterisation of the assumptions underlying the Neo-Gricean tradition:^{2,3}

(2) NEO-GRICEAN MAXIM OF QUALITY (NG-QUALITY)

If the speaker said p , then the speaker believes that p is true.

(3) NEO-GRICEAN MAXIM OF QUANTITY (NG-QUANTITY)

If p and q are both relevant to the conversation, q is more informative than p , and q is among the scalar alternatives of p , then if the speaker believes both are true, the speaker should utter q rather than p .

To illustrate how these maxims interact, consider:

(4) Some of the jewels are missing.

SOME

Assuming the speaker is observing the maxims, a belief-based interpretation of the utterance in (4) is that the speaker does not believe that all the jewels are missing. This inference follows from NG-Quantity: had the speaker believed that all of the jewels are missing, they would have uttered the stronger, more informative statement in (5).

(5) All of the jewels are missing.

ALL

The hearer's reasoning can then proceed in two ways, depending on the common ground and their own assumptions about the speaker's doxastic stance. First, the hearer may simply conclude that the speaker lacks the belief that ALL, as in (6a). Alternatively, if the hearer assumes the speaker is opinionated—i.e., that the speaker holds the belief that ALL or the belief that $\neg$ ALL—they may conclude that the speaker believes $\neg$ ALL, as in (6b).

(6) a. $\leadsto$ the speaker does not hold the belief that ALL is true

b. $\leadsto$ the speaker holds the belief that ALL is false

Importantly, these inferences are about the speaker's beliefs, not about the actual truth of the alternative. While this result is adequate for many purposes, some accounts—both within the Gricean (Gazdar 1979) and grammatical traditions (Fox 2007; Chierchia et al. 2012; Marty 2017;

² See Sperber & Wilson (1986) for another significant development of intention-based pragmatics and implicature derivation, which builds on and diverges from Grice's schema.

³ For expository simplicity, we assume that the modality underlying NG-Quality and NG-Quantity is doxastic, represented by the B -operator. Some authors, however, take the relevant modality to be epistemic, represented by the K -operator, and gloss these inferences accordingly in terms of the speaker's knowledge (e.g., Gazdar 1979 and subsequent work). While this distinction has consequences for the strength of the derived inferences, it does not materially affect the mechanisms for SI derivation presented here, nor the arguments against this family of approaches developed below.

Marty & Romoli 2021)—aim to directly derive the content-level inference that ALL is false. Within the Neo-Gricean tradition, however, a prevailing claim is that SIs are epistemically modalized (Soames 1982; Horn 1989; Sauerland 2004). On this view, Quantity yields weak inferences like (6a), sometimes called *primary implicatures*. Stronger inferences like (6b), referred to as *secondary implicatures*, require an additional “epistemic step” through the assumption that the speaker is opinionated about the alternatives of interest (van Rooij & Schulz 2004; Sauerland 2004; Spector 2007).

As supporting evidence, Sauerland (2004) discusses sentences like (7) where, for a subset of alternatives, the derivation of secondary implicatures must be blocked to ensure consistency, attributing to the speaker ignorance rather than belief in a negated alternative:⁴

- (7) Lupin hid the jewels in the attic or in the basement. A OR B

By NG-Quality, the hearer concludes that the speaker believes A OR B: $B_s(A \text{ OR } B)$. Assuming that the topic of conversation is where Lupin hid the jewels, the relevant scalar alternatives to (7) are as shown in (8) (here and in the following, we use $Alt(S)$ to refer to those alternatives to S that are contextually relevant and more informative than S).

$$(8) \quad Alt(A \text{ OR } B) = \left\{ \begin{array}{ll} \text{Lupin hid the jewels in the attic} & A \\ \text{Lupin hid the jewels in the basement} & B \\ \text{Lupin hid the jewels in the attic and the basement} & A \text{ AND } B \end{array} \right\}$$

By NG-Quantity, the hearer can infer that the speaker does not believe any of these alternatives to be true:

- (9) Primary implicatures: $(\neg B_s(A), \neg B_s(B), \neg B_s(A \text{ AND } B))$

As before, primary implicatures may be strengthened if the speaker is assumed to be opinionated about the truth-value of the members of $Alt(A \text{ OR } B)$. In this case, however, Sauerland (2004) observes that assuming *full* opinionatedness would lead to contradictions: if we were to assume that the speaker is opinionated about each $S' \in Alt(A \text{ OR } B)$, we would conclude $B_s \neg(A)$, $B_s \neg(B)$ and $B_s \neg(A \text{ AND } B)$, in contradiction with the Quality-based implicature $B_s(A \text{ OR } B)$. As Sauerland notes, this suggests that a consistency filter along the lines of (10) is needed to determine which assumptions hearers may entertain.⁵

⁴ For an overview and critical discussion of the range of inferences that plain disjunctions such as (7) may give rise to—including ignorance, exclusivity, and *ad hoc* inferences—see the recent experimental work of Nicolae et al. (2024a; b), Tsvilodub et al. (2024), Marty et al. (2024), and Degano et al. (2025).

⁵ On Sauerland’s 2004 account, the sets of primary (PI) and secondary (SI) implicatures for a sentence S are defined as in (i), where $Alt(S)$ is the set of scalar alternatives S' to S that asymmetrically entail S .

(i) a. $PI(S) = \{\neg B_s(S') \mid S' \in Alt(S)\}$
 b. $SI(S) = \{B_s(\neg S') \mid S' \in Alt(S) \text{ and } B_s(\neg S') \text{ is consistent with } \{B_s(S)\} \cap PI(S)\}$

(10) SAUERLAND'S CONSISTENCY FILTER

For $S' \in \text{Alt}(S)$, the hearer may assume that the speaker is opinionated about S' only if this assumption is consistent with the common ground and previously drawn inferences from NG-Quality and NG-Quantity.

The idea behind (10) is that opinionated speaker assumptions are only licensed if they do not result in attributing contradictory beliefs to the speaker. In the case of (7), this predicts that the hearer may assume that the speaker is opinionated about $A \text{ AND } B$, but not about A or B individually (assuming the speaker is opinionated about A would yield $B_s(\neg A)$ which, together with $B_s(A \text{ OR } B)$, entails $B_s(B)$, contradicting the primary implicature $\neg B_s(B)$; the same logic applies symmetrically to B). Thus, of the primary implicatures above, only $\neg B_s(A \text{ AND } B)$ can be strengthened to a secondary implicature:

(11) Secondary implicatures: $(B_s \neg(A \text{ AND } B))$

This account correctly predicts that the hearer may infer that the speaker believes that not both disjuncts are true, but cannot infer that the speaker believes either disjunct to be false. Combined with the Quality-based implicature, the primary implicatures $\neg B_s(A)$ and $\neg B_s(B)$ yield so-called *ignorance inferences*:

- (12) a. $\rightsquigarrow$ the speaker is ignorant as to whether A
 b. $\rightsquigarrow$ the speaker is ignorant as to whether B

In sum, belief-based accounts hold that cooperative speakers aim to make the most informative contribution compatible with their doxastic state, relative to a given set of alternatives. Hearers, in turn, interpret the speaker's utterance by reasoning about what the speaker most likely believes, based on what they said and what they could have said, but didn't. Sauerland's 2004 proposal—building on work by Soames (1982), Horn (1989; 2004), and others—exemplifies how this reasoning can be formally modelled. Abstracting away from the specifics of disjunction, the broader schema for SI derivation can be synthesized as follows:

1. The speaker uttered S .
2. By NG-Quality, they would not have done so unless they believed S . Thus, $B_s(S)$.
3. By NG-Quantity, they would not have done so if they believed a stronger alternative $S' \in \text{Alt}(S)$. Thus, for all $S' \in \text{Alt}(S)$, $\neg B_s(S')$.
4. For any $S' \in \text{Alt}(S)$, if the opinionated speaker assumption $B_s(S') \vee B_s(\neg S')$ is licensed, and the hearer makes this assumption, they infer $B_s(\neg S')$.

Before turning to Geurts' proposal, one clarification is in order. While Neo-Gricean accounts often streamline or abstract away from the full psychology of intentions—especially in their more

formal incarnations—they largely retain the centrality of belief (or a closely related epistemic notion) as the key truth-directed attitude underlying SIs. Even when these accounts adopt a more algorithmic formulation, belief continues to anchor Quality and to support the derivation of SIs. That said, the extent to which a rich psychological interpretation is assumed varies: some accounts, including the one presented here, treat belief in a relatively schematic way, without committing to a detailed theory of mental representation, while others—particularly in game-theoretic or Bayesian pragmatics—reinterpret belief in terms of credences or probabilistic states. In the general case, however, the derivation of SIs still turns on reconstructing what the speaker takes to be true (or likely), and using that attribution to derive further inferences.⁶ This is precisely where the commitment-based approach departs, by shifting the explanatory focus from mental-state attribution to tracking the normative positions speakers undertake in interaction.

2.3 Commitments and the recasting of Gricean maxims

Geurts (2019b) proposes a significant shift from belief-based models by reconceptualising implicature derivation in terms of *commitments*. This shift is part of a broader move from a psychological model of communication to a more socially grounded one—in Geurts’ own words, a model that treats communication as “coordinated action for action coordination”. On this view, the primary function of face-to-face communication is not to share beliefs but rather to negotiate commitments that facilitate joint action between agents.

A central assumption in Geurts’ proposal is that *every* speech act commits the speaker to the truth of some proposition. Hearers, in turn, interpret an utterance against a background of mutual commitments and shared conversational goals. Importantly, commitments are not psychological attitudes but social and normative ones: unlike private mental states like beliefs, commitments are directed toward and relativized to others, much like obligations and duties. In this framework, a commitment is a three-place relation among two agents and a proposition (Singh 2000): $C_{a,b}(p)$ means that agent a is committed to agent b to act in accordance with p —that is, to act *as if* p is true. This construal foregrounds the normative force of commitment: by committing to p , a undertakes a set of responsibilities, including providing justification for p if challenged, endorsing commitments that follow from p , avoiding commitments incompatible with p , among others.⁷ These normative consequences not only constrains a ’s behaviour, but also entitle b to form expectations and take

⁶ Thanks to an anonymous reviewer for helpful discussion of this point.

⁷ Here and in the following, we assume that the normative powers of commitments are governed by at least three general principles. For any agents a, b : (i) $C_{a,b}(p \text{ and } q)$ iff $C_{a,b}(p)$ and $C_{a,b}(q)$; (ii) if p entails q and $C_{a,b}(p)$, then $C_{a,b}(q)$; and (iii) if $C_{a,b}(p)$, then $\neg C_{a,b}(\neg p)$. Principle (i) reflects the decomposability of conjunctive commitments. Principle (ii) captures the idea that commitment is closed under entailment. Finally, principle (iii) enforces consistency by ruling out incompatible commitments; together with (i), it also precludes commitment to contradictions. Geurts’ Maxim of Integrity, introduced in (13), can be seen as a related constraint governing the overall coherence of commitment sets.

action on the basis of the commitment. Consider (4) again, and suppose that it is uttered by an employee at a jewellery store addressing their boss:

(4) Some of the jewels are missing. SOME

By uttering (4), the employee incurs the commitment $C_{a,b}(\text{SOME})$: they commit to the boss to act as if “some of the jewels are missing” is true. This commitment may carry practical implications. For instance, the employee may now be expected to provide an inventory of the missing jewels, recount events leading up to the theft, report to the police, etc. These expectations may be in force whether or not the speaker is fully aware of them at the time of utterance. The boss, in turn, is entitled to act on the same proposition, would he wish to do so—e.g., by launching an internal investigation, contacting authorities, etc. In such cases, the proposition expressed by (4) becomes a mutual commitment—a constraint shaping the subsequent behaviour of both agents and governing their interactions.

This perspective raises a natural question when applied to conversational implicature: how can a socially directed commitment to p license the inference that the speaker is *personally* committed to p ? It is a well-established feature of speech acts that they typically license speaker-oriented inferences of *sincerity* (Searle 1969; Searle & Vanderveken 1985): if someone asserts (4), we infer that they take this to be true; if they promise to go to the police, we infer they intend to do so; if the boss orders them to file a complaint, we infer that the boss intends the employee to do so, and so on. So why, and how, do other-directed commitments give rise to inferences about commitments that govern the agent’s own conduct? This is, in effect, the challenge of recasting the maxim of Quality within a commitment-based framework.

The difficulty is that the first clause of the Maxim—“Do not say what you believe to be false”—appeals to private mental states and thus resists straightforward reformulation in terms of social commitments. Geurts argues, however, that Quality inferences can be captured in terms of *private commitments*, that is, commitments an agent undertakes toward herself (Clark 2006; Geurts 2018). In essence, private commitments serve the same coordination function as social ones, but at the intra-personal level: just as agents undertake social commitments to coordinate their actions with others ($C_{a,b}(p)$, where $a \neq b$), they undertake private commitments to coordinate their own actions ($C_{a,a}(p)$). Crucially, whereas social commitments primarily constrain public behaviour, private commitments constrain the agent’s conduct across the board, including self-directed action.⁸

⁸ The notion of private commitments is further motivated by communicative settings in which speaker and hearer coincide, as in self-talk. As Geurts (2018) argues, self-directed speech is puzzling if communication is conceived as mere information exchange, since exchanging information with oneself appears pointless. By contrast, if communication is understood as the negotiation of commitments, self-talk becomes intelligible: undertaking commitments to oneself is no less important than undertaking them toward others.

For present purposes, the key point is that private commitments can play much of the theoretical role traditionally assigned to the *normative aspects* of belief and intention (see also Brandom 1997). For instance, if the employee's private conduct is guided by the premise that some of the jewels are missing, then the employee is privately committed to that proposition—and, in that case, believes it; if the boss's private conduct is guided by the premise that she will call the police, then the boss is privately committed to doing so—and thereby intends to do so. Importantly, the notion of private commitment employed here is both more general and more selective than the notion of belief typically assumed in Gricean and Neo-Gricean approaches. It is more general because it extends beyond paradigmatic doxastic attitudes to cover any case in which an agent undertakes to regulate their own conduct in accordance with a proposition. It is more selective because it isolates the action-guiding, normative dimension of belief, abstracting away from its causal-psychological aspects. In this sense, private atelic commitments capture what we might call the *normative profile* of belief—what an agent ought to rely on, and is answerable, for in reasoning—rather than its underlying psychological realisation.⁹

Returning to Quality inferences, Geurts's account starts from the assumption that successful action coordination requires agents to avoid undertaking conflicting commitments. Cooperative agents, therefore, are expected to obey the following maxim:

(13) MAXIM OF INTEGRITY

If agent a is committed to b to act on p , a must not commit to c to act on $\neg p$.

Technically, if $C_{a,b}(p)$, then $\neg C_{a,c}(\neg p)$.

Particularly relevant is the application of this maxim to the case where $c = a$. Since the maxim holds for any a , b , and c , it follows that agents should not publicly commit to what they are privately committed against, (14). Concretely, if the employee believes that no jewels are missing, he shouldn't tell his boss that some of them are. The fact that he did utter (4) implicates—by Sincerity—that he is not privately committed to $\neg$ SOME.

(14) SINCERITY (sub-maxim of Integrity)

If agent a is socially committed to p , a must not be privately committed to $\neg p$.

Technically, if $C_{a,b}(p)$, then $\neg C_{a,a}(\neg p)$.

Sincerity implicatures, however, are weaker than what we usually infer: they explain why we accept the consequence in (15a) but not why we also readily accept the one in (15b).

⁹ A normative conception of belief along these lines is found, for instance, in Lewis (1969). In broadly Lewisian terms, believing p amounts to adopting a stance one is rationally permitted or required to hold given one's evidence or reasons: $B(p)$ may be modelled as sufficiently high credence in p , where credences are constrained by norms of rationality (see also Skyrms 1987; Joyce 1998). On this view, belief tracks what a rational agent ought to rely on—and is answerable for—in reasoning. This normatively construed notion of belief is compatible with—and in paradigmatic cases of assertions coincides extensionally with—the notion of private commitment introduced here.

- (15) If the employee tells his boss that some jewels are missing,
- a. he doesn't believe that no jewels is missing.
 - b. he believes that some of them are missing.

The distinction between (15a) and (15b) is reminiscent of the distinction between primary and secondary implicatures and Geurts proposes to apply a similar analysis here: Sincerity implicatures of the form $\neg C_{a,a}(\neg p)$ are strengthened to $C_{a,a}(p)$ in contexts where it can be assumed that the speaker is privately committed to p or $\neg p$, that is, when $C_{a,a}(p) \vee C_{a,a}(\neg p)$ holds. This auxiliary premise—let's call it the *privately committed speaker* assumption—plays a role analogous to the *opinionated speaker* assumption in Neo-Gricean accounts of Quantity. Given $\neg C_{a,a}(\neg p)$ and $C_{a,a}(p) \vee C_{a,a}(\neg p)$, it follows that the speaker is privately committed to p , $C_{a,a}(p)$. On this view then, Quality inferences emerge as a by-product of Sincerity combined with auxiliary committed speaker assumptions.¹⁰

More importantly for our present concerns, Geurts suggests that just as Quality implicatures can be derived without direct reference to mental states, so too can Quantity implicatures. He sketches a derivational schema for SIs that preserves the key steps of belief-based models while replacing beliefs with commitment-based premises:

1. In uttering (4), the speaker committed to the hearer to act on SOME.
2. The speaker could have taken a stronger commitment by uttering the statement in (5).
3. The most likely explanation is that the speaker doesn't have sufficient evidence to be committed to act on ALL.
4. If it can be assumed that the speaker has enough evidence to act on either ALL or $\neg$ ALL, we infer that the speaker is committed to act on $\neg$ ALL.

At a general level, this approach allows us to remain neutral on the metaphysics of beliefs and their psychological realisation. Importantly, it does not require abandoning the idea that speech acts express mental states. On the contrary, the central role of private commitments is fully compatible with the view that, in the general case, speech acts do express beliefs and intentions. What the approach proposes, rather, is that the relevant pragmatic inferences can be derived by appealing directly to the normative structure of commitments—what agents are publicly and privately answerable for—without requiring mediation by assumptions about underlying psychological states. In this sense, mental-state attribution is no longer treated as a prerequisite

¹⁰ This aspect of Geurts's proposal is revisited in Section 4.1. In brief, I argue that deriving Quality inferences in part from defeasible auxiliary assumptions fails to capture their resistance to cancellation and reinforcement. I suggest instead that the link between $C_{a,b}(p)$ and $C_{a,a}(p)$ is more direct than Geurts assumes and follows from the reflexive, self-involving nature of social commitments in cooperative settings. As this issue is not central to the present discussion, I set it aside for now.

for communication. This approach can thus be seen not as competing with belief-based accounts, but as recasting part of their explanatory role in explicitly normative terms.

At a more specific level, Geurts provides a proof of concept that Quality implicatures can be derived within a commitment-based framework and sketches a parallel path for Quantity implicatures. This opens the door to a commitment-theoretic account of SIs, one that retains Grice's architecture while grounding it in the normative domain of commitments. However, Geurts' reconstruction of Quantity-based reasoning remains programmatic and leaves several important questions open: How is Quantity to be recast in this framework? What is the strength of the implicatures so derived? What assumptions are required to support them? And how, exactly, are social and private commitments related in SI derivation? Addressing these questions is essential to determining whether the shift pursued here yields genuine explanatory and empirical advantages over more familiar approaches. This is the aim of the following sections.

3 Toward a commitment-based account of SIs

This section outlines a conservative implementation of Geurts's commitment-based schema for SI, relying exclusively on Sincerity and auxiliary committed speaker assumptions. This preliminary account serves as a useful starting point for rethinking SI derivation in commitment-theoretic terms. As we shall see, however, it faces both conceptual and empirical challenges when it comes to the interplay between social and private commitments. These limitations will motivate a refined proposal, developed in the next section.

3.1 Laying the groundwork

Consider (4) again, alongside the minimally different but stronger statement in (5):

- | | |
|---|------|
| (4) Some of the jewels are missing. | SOME |
| (5) All of the jewels are missing. | ALL |

As discussed, the speaker's choice to utter SOME rather than ALL—in a context where both are relevant—prompts the addressee to consider why ALL was not chosen. These reflections form the basis of scalar reasoning, which, in a commitment-based framework, must be (re)constructed in terms of normative relations between agents and propositions. Thus, to set the stage, we begin by addressing two foundational questions: (1) How are the alternatives entering Quantity-based reasoning determined in a commitment-based model? (2) What kinds of implicatures arise from such reasoning?

Like other models, a commitment-based account of SIs requires a theory of alternatives—an account of what counts as a scalar alternative to a given sentence—addressing, among other

challenges, the so-called “Symmetry Problem” (a.o., Kroch 1972; Fox 2007; Katzir 2007; Fox & Katzir 2011; see Breheny et al. 2018 for an overview). For our purposes, any extant theory of alternatives will do. Virtually, all such theories agree on the minimal characterisation of $Alt(S)$ given below:

- (16) ALTERNATIVES: $S' \in Alt(S)$ only if (i) S' is a scalar alternative to S , (ii) the proposition p' expressed by S' asymmetrically entails the proposition p expressed by S , and (iii) p' is relevant in the utterance context of S .

This characterisation, while minimal, is general enough for our purposes. In particular, it can apply, beyond assertions, to any speech act capable of generating commitments, including directives, questions, and so forth. Consider for instance the following directive:

- (17) Hide some of the jewels in the attic.

Here, the speaker (the thief) issues a directive to the addressee (the accomplice), thereby committing to act as if the addressee will carry out the relevant action. This *telic* commitment differs from the *atelic* commitment in (4), in that the proposition it targets—*that the accomplice hides some of the jewels in the attic*—is one whose truth depends on the addressee’s future actions. Still, by uttering (17), the speaker undertakes a committing to that proposition and, by (16), this commitment stands in a scalar relation to the stronger directive in (18). Therefore, scalar enrichment is expected to arise here as well and to proceed in a way analogous to the enrichment observed for (4).

- (18) Hide all of the jewels in the attic.

Turning to the second question, recall that, in this framework, an utterance of (4) is primarily an act of establishing a social commitment: the speaker commits to the addressee to act on SOME, nothing more. On Geurts’ view, it takes a Sincerity assumption to derive $\neg C_{a,a}(\neg \text{SOME})$ from $C_{a,b}(\text{SOME})$, and an additional assumption of speaker commitment to reach the private stance $C_{a,a}(\text{SOME})$. In this setup, scalar reasoning could proceed as follows: had the speaker uttered ALL, they would have sought to establish $C_{a,b}(\text{ALL})$; since they didn’t, they implicate that they are not committed to act on ALL, $\neg C_{a,b}(\text{ALL})$. This invites a recasting of Quantity on which, for any $S' \in Alt(S)$, if a can commit to b to act on the propositions expressed by both S and S' , a should utter S' rather than S . With this in mind, we can propose a provisional mechanism for generating primary implicatures:¹¹

¹¹ I will propose in Section 4 a refinement of (19), based on further considerations about the relation between public and private commitments (see 3.3). The current formulation, however, already captures important results (presented below) that will be preserved in the revised account.

(19) For any sentence S uttered by agent a to agent b :

$$PI(S) = \{\neg C_{a,b}(p') \mid p' \text{ is the proposition expressed by } S', \text{ where } S' \in Alt(S)\}$$

Applied to (4), this mechanism yields the implicature $\neg C_{a,b}(ALL)$. This implicature is weak, but it is nonetheless correct and aligns with the fundamental idea behind Quantity: drawing attention to more committal propositions the speaker could have expressed but didn't. Importantly, this marks a significant departure from belief-based models: while NG-MQ would yield here the inference that the speaker does not believe ALL , the negation of a social commitment leaves the speaker's private commitments entirely open. In other words, on this proposal, a cooperative speaker may well believe the stronger proposition but not be disposed to share that belief with the addressee. This, I argue, is a desirable result: it captures a pervasive phenomenon in real-world discourse, which I will call *semi-opacity*, where the speaker's private commitments are only partially revealed due to other constraints limiting their disposition to share the whole of their private commitments with other agents. To illustrate, consider the following examples:

- (20) **Guarded speech 1:** Detective Sholmes has gathered strong evidence that Lupin stole all the jewels. The detective personally believes this to be true and is acting on that belief (e.g., searching for the full set), but lacks enough proof to make that belief publicly assertable. In handing off the case file to a colleague, he might appropriately say: "Lupin stole some of the jewels."
- (21) **Guarded speech 2:** An FBI agent is asked about the transfer of a high-profile prisoner. The question is whether the prisoner will be extradited or remain on U.S. soil, and if so, where exactly. The agent knows the prisoner is being transferred to the Huntsville Unit in Texas, but he may only say: "The prisoner will be transferred to a facility somewhere in Texas."

In both examples, the speaker could have uttered a stronger statement that they believe to be true. Yet their choice of a weaker statement is perceived as fully cooperative. These statements also sound natural, meaning that no inference arises attributing to the speaker a lack of doxastic commitment to the stronger alternative. On a neo-Gricean view à la Sauerland, this would require assuming that NG-MQ is deactivated in such contexts. The commitment-based account offers another explanation: Quantity does not force any conclusion about the speaker's private commitments.

More broadly, these examples illustrate that speakers may refrain from committing to a stronger proposition not because they disbelieve it, but because they are constrained from endorsing it socially. In (20), for instance, Sholmes might avoid asserting the *all*-alternative in order to preserve his colleague's impartiality; alternatively, the motivation might be more reputational than methodological: he may be seeking to protect his credibility and avoid

potential blame for endorsing a conclusion that could later be proven incorrect.¹² In (21), the speaker is more likely bound by institutional or tactical constraints—revealing the exact transfer location could compromise the operation. In both cases, then, the choice to withhold a stronger commitment stems not from epistemic limitations but from other pragmatic pressures being in play.

To summarise, on this implementation of Geurts’ proposal, the implicatures generated by Quantity reflect only what the speaker is *not socially committed to*, leaving their private commitments—i.e., their beliefs, in the case of atelic commitments—open. This feature of the model avoids over-ascribing ignorance or disbelief to the speaker, and enables a more nuanced interpretation of guarded speech. More generally, it allows us to account for cooperative communication in semi-opaque contexts where agent’s disposition to share private commitments is affected by other constraints. Whereas other accounts might require positing that Quantity is deactivated in such contexts, the commitment-based model preserves its operation while recognising that there are often principled limits on what agents may be disposed to commit to in front other agents.

3.2 From social to private commitments

We have hypothesised so far that Quantity operates at the level of social commitments. Specifically, we have assumed that the effects of Quantity are captured via the (provisional) mechanism in (19) which, in a case like (4), yields the implicature $\neg C_{a,b}(\text{ALL})$. In the general case, however, hearers appear to draw inferences about the speaker’s private commitments. For instance, they may readily accept the consequence in (15a) or the stronger one in (15b):

- (22) If the employee told his boss that some jewels are missing,
- a. he doesn’t believe that all of the jewels are missing.
 - b. he believes that not all of the jewels are missing.

So how do we get from the level of shared commitments between *a* and *b* to that of private commitments between *a* and themselves? In principle, this can be achieved using the mechanisms already introduced. In particular, Geurts’ notion of Sincerity, linking social and private commitments, can be used here too. The key observation is that, while an absence of commitment like $\neg C_{a,b}(\text{ALL})$ cannot trigger Sincerity, its stronger form, $C_{a,b}(\neg \text{ALL})$, can. This stronger form can be derived from the Quantity implicature $\neg C_{a,b}(\text{ALL})$ if we assume, for instance, that *a* is socially committed to ALL or to $\neg \text{ALL}$:

¹² My goal here is not to provide a taxonomy of the reasons why private commitments may remain partially opaque without conversational oddity. Rather, it is to show that this phenomenon is both more common than typically acknowledged and well captured by the commitment-based approach.

$$(23) \quad C_{a,b}(\text{ALL}) \vee C_{a,b}(\neg\text{ALL})$$

SOCIALLY COMMITTED SPEAKER

This assumption mirrors the assumption of privately committed speaker discussed earlier; the only difference lies in the level at which the commitment holds. In effect, (23) merely rules out the possibility that a is non-committal to b about ALL. Given $\neg C_{a,b}(\text{ALL})$, the first disjunct in (23) is excluded, and $C_{a,b}(\neg\text{ALL})$ follows. From there, Sincerity may apply, licensing the inference $\neg C_{a,a}(\text{ALL})$. This inference can then be strengthened to $C_{a,a}(\neg\text{ALL})$ in a way analogous to Geurts' proposal for Quantity implicatures—namely, by assuming that a is *privately* committed to ALL or to $\neg\text{ALL}$:

$$(24) \quad C_{a,a}(\text{ALL}) \vee C_{a,a}(\neg\text{ALL})$$

PRIVATELY COMMITTED SPEAKER

Thus, the resulting account is one on which both types of speaker assumptions—social and private—play a critical role in deriving the target inference and Sincerity a pivotal role in bridging these two levels. The socially committed speaker assumption strengthens the initial Quantity implicature to a point where Sincerity becomes applicable and Sincerity, in turn, enables inferences about private commitments, which may be bolstered by the privately committed speaker assumption. Taken together, these mechanisms can account for why, in uttering (4), a speaker may be taken to socially commit to $\neg\text{ALL}$ and why that social commitment may, in turn, be interpreted as reflecting a private one.

Despite these good results, this account raises two independent concerns. The first pertains to the socially committed speaker assumption, which is crucial for triggering Sincerity. While this assumption structurally mirrors the privately committed speaker assumption, it cannot be justified in the same way. On the one hand, the private assumption can be motivated by reasoning about the evidence the speaker plausibly has access to—his “expertise”, so to speak. The evidential basis for its social counterpart, by contrast, is unclear: no matter how compelling the speaker’s expertise about p' is, it does not tell us, absent further assumption, whether the speaker is committed to act on p' or $\neg p'$ vis-à-vis the addressee. What, exactly, in the context justifies this premise? As the account currently stands, whether the premise holds seems to depend entirely on how plausible it is to attribute that disposition to the speaker in the given context.¹³

The second concern relates to the direction of derivational dependency between social and private commitments. In the current model, inferences about a speaker’s private commitments must be licensed by prior inferences about their social commitments. Thus, rather than allowing our reasoning about a ’s private commitments to ground inferences about their disposition to share those commitments with b , the current account reverses the dependency: it makes conclusions

¹³ One could hypothesise that such premisses are generated by *default*: a speaker uttering S is presumed to take a determinate stance vis-à-vis their addressee on the alternatives in $\text{Alt}(S)$, unless such a stance is incompatible with the common ground. But without an independent rationale for this default, some explanation of what conversational norm or pragmatic pressure supports it, this move seems *ad hoc*.

about the speaker's private stance contingent on assumptions about their social ones. This feature seems counter-intuitive and questions the adequacy of using Sincerity as the sole or main bridging mechanism. In the following, we show that it also restricts the explanatory scope of the account, which struggles to accommodate cases where hearers may infer that the speaker lacks a private commitment.

3.3 Lack of private commitments

In addition to conveying commitments beyond what is said, speakers may also convey their *lack* of commitment to certain propositions. Such cases typically yield the inference that the speaker is not in a position to endorse a more committal alternative—both socially and privately. One way in which this arises in spoken discourse is through the use of certain intonational contours. Consider the two possible prosodic realisations of (4) below:¹⁴

- (25) a. Some of the jewels are missing.
 H* (H*) L-L%
 ↗ *the speaker believes that not all of the jewels are missing*
- b. Some of the jewels are missing.
 L + H* (H*) L-H%
 ↘ *the speaker believes that not all of the jewels are missing*

The contour in (25a) typically supports the inference from SOME to $\neg$ ALL. By contrast, the rise-fall-rise (RFR) contour in (25b) indicates that the speaker is giving a non-resolving answer to a contextually salient question (Constant 2012), suggesting instead uncertainty on the speaker's part as to whether ALL (Ward & Hirschberg 1985; Pierrehumbert & Hirschberg 1990; Constant 2012; see also Tomlinson & Ronderos 2021 for experimental data). Crucially, as Constant (2012) shows, this inference is not entailed by (the semantics of) the RFR contour itself. In particular, this contour can also be felicitously used by a speaker opinionated about ALL, as illustrated in (26) (adapted from Constant 2012, ex.67).

- (26) [Some_{L+H*} of the jewels are missing_{L-H%}], but the diamond pendant is still there.

For our purposes, the contrast in (25) teaches us that a speaker may utter (4) to convey non-commitment with respect to ALL—both socially and privately—and be understood accordingly. This poses a challenge for the current account. As it stands, this account may derive the private non-commitment to ALL (through Sincerity) only if the social commitment to $\neg$ ALL is first derived. This inferential path is clearly unwarranted here. Alternatively, the hearer may suspend the

¹⁴ Annotations follow the Tones and Break Indices (ToBI) system for American English (Silverman et al. 1992; Beckman & Ayers Elam 1997), using labelling conventions developed by Pierrehumbert and colleagues (Pierrehumbert 1980; Beckman & Pierrehumbert 1986).

socially committed speaker assumption that $C_{a,b}(\text{ALL}) \vee C_{a,b}(\neg \text{ALL})$ to accommodate the non-committal stance suggested by the RFR contour. But if this assumption doesn't hold, Sincerity can no longer be used to reach the speaker's private stance and so (25b) is left unaccounted for.

As discussed in Section 2.2, inferences to a speaker's lack of commitment also arise in certain environments in the absence specific intonational cues, like disjunctive statements:

(7) Lupin hid the jewels in the attic or in the basement. A OR B

To see what the current account predicts for such cases, consider first the set of primary implicatures generated by (19):¹⁵

(27) $\neg C_{a,b}(A), \neg C_{a,b}(B), \neg C_{a,b}(A \text{ AND } B)$

Assuming that committed speaker assumptions are checked for consistency (cf. (10)), only one of these inferences—namely, $\neg C_{a,b}(A \text{ AND } B)$ —can be strengthened in the way described earlier. This strengthening yields the social commitment $C_{a,b}\neg(A \text{ AND } B)$, and in turn, the corresponding private commitment $C_{a,a}\neg(A \text{ AND } B)$. No such strengthening is available for the other two primary implicatures: $C_{a,b}(A \text{ OR } B)$, together with the implicatures in (27), already entails that a is socially non-committal with respect to A and to B individually. Thus, the present account predicts the following inferences:

(28) a. $C_{a,b}\neg(A \text{ AND } B)$ and $C_{a,a}\neg(A \text{ AND } B)$
 b. $\neg C_{a,b}(A) \wedge \neg C_{a,b}(\neg A)$
 c. $\neg C_{a,b}(B) \wedge \neg C_{a,b}(\neg B)$

Note that none of these inferences can be further strengthened within the present account. In particular, we cannot derive the inference that a 's social non-committal stance toward A and B also reflects a private one, meaning that the inferences in (28) are compatible with both propositions in (29) being true. This effectively predicts that an utterance of (7) is consistent with a being privately committed to act on A and $\neg B$, or conversely, on B and $\neg A$. Thus, on this analysis, a 's non-committal stance toward the individual disjuncts may be purely a social posture, constraining only their public actions.

(29) a. $C_{a,a}(A) \vee C_{a,a}(\neg A)$
 b. $C_{a,a}(B) \vee C_{a,a}(\neg B)$

While this outcome may be desirable for disjunctive statements in semi-opaque contexts (we discuss these cases further in Section 4), it does not explain why, in casual conversation, utterances

¹⁵ As before, we assume that the scalar alternatives to a sentence like (7) are the individual disjuncts and their conjunction, noted A , B , and $A \text{ AND } B$, respectively; that is, $Alt(A \text{ OR } B) = \{A, B, A \text{ AND } B\}$.

of (7) may readily give rise to the inference that the speaker is wholeheartedly ignorant as to which disjunct is true—that is, that the speaker lacks private commitments regarding either disjunct. We conclude, then, that such cases leave us with an open issue.

3.4 Intermediate summary

We outlined a possible implementation of Geurts’s proposal relying exclusively on Sincerity and committed speaker assumptions. The resulting account succeeds in some respects. Most notably, through the interplay of social and private commitments, it helps account for how scalar reasoning unfolds in semi-opaque contexts. Yet several limitations emerged. Conceptually, the assumptions underlying the notion of a “socially committed speaker” remains insufficiently motivated; empirically, the hypothesis that private commitments are derived from social ones fails to derive the genuine lack of private commitments. These observations suggest that the principle we assumed to mediate between social and private commitments—Sincerity—is not well-suited to the dynamics of SI derivation.

4 A (revised) commitment-based account of SIs

This section develops a revised account aimed at meeting the challenges identified in Section 3. It seeks to preserve the strengths of the current approach, particularly its treatment of semi-opaque contexts, while addressing the conceptual and empirical gaps we identified. In doing so, it introduces a new bridge between social and private commitments—Transparency—better aligned with the demands of scalar reasoning. Before introducing this notion, we first return to the more fundamental issue of how the maxims of Quality and Quantity are to be properly integrated within a commitment-based model.

4.1 Remarks on Quality

As discussed in Section 2.3, Geurts (2019a) relates Quality to the observation that conflicting commitments undermine action coordination: when two incompatible bids for coordination are issued, one is bound to fail. Cooperative agents should therefore avoid such conflicts. This idea is formalized as the Maxim of Integrity, from which the sub-maxim of Sincerity follows (see (13) and (14)). On Geurts’ proposal, however, Sincerity alone does not yield full-fledged Quality inferences: these arise only in contexts where additional committed-speaker assumptions can be made. As we see it, the reliance on such assumptions—by definition optional and defeasible—raises both conceptual and empirical concerns.

The empirical concern is that, if Quality inferences are mediated by defeasible assumptions, we would expect them to display the characteristic flexibility of other implicatures. Yet this expectation is not borne out. As noted early on by Levinson (1983) and Horn (1984), Quality

inferences lack some of the defining properties of SIs in resisting both cancellation and reinforcement:¹⁶

(30) **Cancellation: Moore's paradox effects**

- a. Lupin stole the jewels, though #I don't know whether he did.
- b. Lupin stole the jewels, though Sholmes doesn't know that he did.

(31) **Reinforcement: Redundancy effects**

- a. Lupin stole the jewels and #I know it.
- b. Lupin stole the jewels and Sholmes knows it.

Intuitively, the anomaly of the (a)-sentences stems from the fact that the 1st person attitude expressed in the second clause is either contradictory or redundant with the inference, triggered by the first clause, that the speaker is vouching for the truth of the proposition *Lupin stole the jewels*. No such effect arises in the (b)-sentences, where the relevant attitudes are attributed to a third party. This pattern is unexpected on Geurts' account. Even if committed-speaker assumptions are treated as strong defaults, speakers should in principle be able to suspend them when doing so would preserve coherence.

The conceptual concern is that, as it stands, Geurts' Maxim of Sincerity appears too weak to secure successful joint action. While it rules out committing to p while being privately committed to $\neg p$, it still allows an agent to commit to p while remaining privately agnostic or indifferent with respect to p . This consequence is problematic: if p is not integrated into the agent's private commitment set, the success of joint action is jeopardised by the risk of misalignment between the agent's social commitment and their personal standpoint. More generally, it is difficult to see how committing to act in accordance with p could be sustained unless p is effectively available as a premise in the agent's own action and deliberation. From this perspective, the gap between $C_{a,b}(p)$ and $C_{a,a}(p)$ left open by Sincerity seems to threaten the practical coherence of commitments.

The proposal advanced here is that this gap should not arise in the first place: for a social commitment to be sustainable and fulfil its coordinating role, it must be privately actionable. Accordingly, social commitments, though directed toward others, must be self-involving: in committing to act on p , a cooperative agent thereby incurs a requirement to have p as a private stance. On this view, the transition from social to private commitment is not mediated by auxiliary

¹⁶ For comparisons, cancelation and reforcability of Quantity-based inference are illustrated in (i). The second clause in (ia) cancels the potential *not-all* SI associated with *some*, while the second clause in (ib) makes it explicit. No Moore's paradox or redundancy effects arise in these cases.

(i)	a. Lupin stole some of the jewels, in fact he stole them all.	Cancellation
	b. Lupin stole some of the jewels, but he didn't steal them all.	Reinforcement

assumptions, but follows from the structure of commitment itself. This idea can be captured by the following Reflexivity principle:¹⁷

(32) REFLEXIVITY (SELF-ALIGNMENT)

Undertaking a social commitment to p places agents under a normative requirement to have the corresponding private commitment.

Technically, if $C_{a,b}(p)$, then $C_{a,a}(p)$.

This proposal has two immediate advantages. First, it resolves the empirical issue: if the link between $C_{a,b}(p)$ and $C_{a,a}(p)$ is partly constitutive of the act of committing itself, the resistance of Quality inferences to cancellation and reinforcement is no longer surprising. Second, it strengthens the conceptual foundations of the framework by ensuring that commitments are sustainable from the agent's own perspective.

The upshot of these remarks is thus partly negative: deriving Quality inferences from Sincerity together with auxiliary assumptions seems inadequate. In this respect, our proposal departs from Geurts' and weakens further the role of Sincerity in conversational reasoning.¹⁸ However, it remains aligned with the rationale underlying Geurts' account: Reflexivity not only rules out cases in which an agent is socially committed to p while privately committed to $\neg p$, but—given standard consistency constraints—it also prevents cooperative agents from undertaking conflicting commitments. Thus, Reflexivity can be grounded in the same coordination-based logic as Integrity. Its formulation also preserves the empirical scope of Geurts' original proposal. Whereas traditional Gricean formulations of Quality are closely tied and tailored to assertions, Reflexivity applies wherever a commitment is undertaken and generalizes across different types of speech act, covering a broad range of inferences corresponding roughly to Searle's 1969 "sincerity conditions" (see also Martinich 1980, Searle & Vanderveken 1985; see Geurts 2019b for discussion).

This advantage becomes particularly clear in the case of institutional or conventional speech acts, where the speaker's mental state is largely irrelevant to the felicity of the act. Consider the following example:

(33) JUDGE: I find the defendant, Arsène Raoul Lupin, innocent of all burglary charges.

¹⁷ Bart Geurts pointed out to me that the contrasts in (30)–(31) might be captured by a weaker linking principle on which undertaking a social commitment to p only *socially* commits the agent to having the corresponding private commitment, that is, if $C_{a,b}(p)$, then $C_{a,b}(C_{a,a}(p))$. On this view, what is added is a higher-order social commitment, without requiring that the corresponding private commitment actually obtain. The difference with (32) thus turns on where the requirement is located: in the agent's own commitment set, ensuring intra-personal guidance (self-alignment), or solely in the public score, ensuring inter-personal answerability (public accountability). The weaker formulation may therefore be attractive for accounts that treat insincere speech or bullshitting as fully regular, rather than normatively defective. I leave a more detailed comparison of these options for future work.

¹⁸ Thanks to an anonymous reviewer for helpful discussion of this point.

In such cases, the felicity and success of the speech act do not depend on the speaker’s beliefs or intentions. Regardless of what the judge believes, the utterance brings about an immediate normative change: a verdict is issued. Still, the act places the speaker under a commitment to act in accordance with its content. Accordingly, a judge who utters (33) is thereby committed to act as if *Lupin is innocent*, and—by Reflexivity—this public commitment triggers a corresponding requirement of private alignment.

4.2 Quantity revised

Turning now to Quantity, let us consider the theory-neutral formulation of the maxim proposed by Marty et al. (2022):

(34) QUANTITY (adapted from Marty et al. 2022)

If p and q are both relevant to the conversation, q is more informative than p , and q is among the alternatives of p , then if a speaker is in a position to communicate both, the speaker should utter q rather than p .

This formulation introduces the notion of “being in a position to communicate”. The question, for a commitment-based account, is how this notion should be understood in terms of agents’ commitments. I propose the following definition:¹⁹

(35) POSITION TO COMMUNICATE

An agent a is in a position to communicate p to an agent b only if a is privately committed to act on p and disposed to share that commitment with b .

Concretely: $PtC_{a,b}(p) = C_{a,a}(p) \wedge D_a(C_{a,b}(p))$.

On this definition, “being in a position to communicate” is a three-place relation among two agents (a , b) and a proposition (p), comprising two components, a private commitment to p and a disposition to share p . The first ensures compliance with the Reflexivity principle in (32): an agent is in a position to communicate p only if they could effectively do so while observing the Reflexivity requirement associated with the relevant commitment. The second should be understood as a *responsive-recognitive disposition* in Brandom’s sense (Brandom 1983). To be disposed to communicate p is, in this sense, to be disposed to undertake the responsibility to justify p if challenged, to acknowledge entitlements to commitments inferentially downstream from p , and to navigate the normative landscape in which p is embedded. Accordingly, whether

¹⁹ In the case of assertions—and more specifically of constatives—private commitments can often be equated with beliefs, so that $C_{a,a}(p)$ may be replaced by $B_a(p)$ without consequence. This simplification, however, does not generalise. Consider, for instance, directives such as (17), “Hide some of the jewels in the attic”. Here, the relevant scalar inference is not naturally cast in terms of belief, but in terms of intention: the speaker intends that the addressee not hide all of the jewels in the attic. A belief-based formulation is therefore not well suited to capture such inferences, whereas in commitment-theoretic terms they are naturally rendered as private commitments.

an agent is disposed to communicate p depends on what the relevant community counts as appropriate justifications for p , what inferences it treats p as licensing and the norms governing its social practices (e.g., institutional codes of conduct, cultural expectations in interpersonal exchanges, politeness).

On this view, by communicating p , a cooperative agent actualises their disposition to commit to p before the addressee, thereby disclosing a private commitment. Conversely, if an agent refrains from communicating p in a context where p is relevant, this warrants the inference that the agent either lacks the corresponding private commitment or is not disposed to share it with their addressee. Thus, the primary implicatures predicted by the present commitment-based treatment of Quantity are characterised as follows:

(36) CB-QUANTITY

For any sentence S uttered by agent a to agent b :

$$PI(S) = \{\neg PtC_{a,b}(p') \mid p' \text{ is the proposition expressed by } S' \text{ and } S' \in Alt(S)\}$$

Given a set of alternatives, CB-Quantity enjoins speakers to select the strongest proposition among those to which they are both privately committed and disposed to share with the addressee. Applied to an utterance of “Some of the jewels are missing” (cf. (4)), CB-Quantity yields the inference that the speaker is not in a position to communicate ALL—that is, either the speaker is not privately committed to ALL, or, if they are, they are not disposed to share that commitment with their addressee:

$$(37) \quad \neg PtC_{a,b}(ALL) = \neg C_{a,a}(ALL) \vee \neg D_a(C_{a,b}(ALL))$$

This reconceptualisation of Quantity both echoes and departs from the account sketched in Section 3. Like the earlier account, it leaves the speaker’s private commitments regarding p' undetermined: the implicature in (37) does not, by itself, settle a ’s private stance toward ALL. However, it diverges from the earlier account in a crucial respect. Rather than merely signalling the absence of a social commitment, (37) discriminates between two distinct explanations for why the speaker did not select the stronger alternative. As we show next, these two possibilities map onto two distinct pathways for scalar reasoning.

4.3 Enter Transparency assumptions

In the present framework, the success of communication is ultimately tied to agents’ ability to coordinate action through communication itself. Although such coordination can, in principle, rely solely on social commitments, this mode of engagement remains fragile: if public and private actions diverge, joint action becomes inefficient. As argued above, the reflexive, self-involving nature of commitment helps mitigate this risk by promoting alignment between inner stance and communicative behaviour.

I now propose that cooperative agents often operate under further regulative expectations by treating the disclosure of private commitments as a predictable feature of communicative behaviour. I refer to such expectations as “Transparency” assumptions:

(38) TRANSPARENCY ASSUMPTION

a has a transparency commitment to b relative to p , notated $Trans_{a,b}(p)$, if and only if, for any $p' \in \{p, \neg p\}$, if a is privately committed to p' , then a is disposed to commit to b to act on p' .

That is, if a is privately committed to p or to $\neg p$, then—under Transparency— a is presumed to be disposed to share that commitment with b . Importantly, these are not assumptions of constant or unconditional disclosure. Rather, they concern an agent’s dispositional openness with respect to private commitments that are relevant to the coordination problem at hand. Their plausibility is therefore context-sensitive: it depends on the norms, expectations, and practical pressures governing a given interaction (e.g. social roles, institutional constraints, sensitivity of the topic, conversational stakes). Accordingly, Transparency assumptions are defeasible. They may be adopted, suspended, or revised dynamically in the course of pragmatic reasoning as interlocutors assess the constraints under which the exchange unfolds. Consider, for instance, the following question:

(39) Have you ever smoked marijuana?

Within the present framework, drawing on Krifka (2015), such a question can be analysed as a request that the addressee commits to one of two propositions: *I have* (YES) or *I have never smoked marijuana* (NO).²⁰ In informal, low-stakes settings, the presumption of Transparency may reasonably hold, yielding the following expectation (where b is the questioner and a the respondent):

(40) $[C_{a,a}(\text{YES}) \rightarrow D_a(C_{a,b}\text{YES})] \wedge [C_{a,a}(\text{NO}) \rightarrow D_a(C_{a,b}\text{NO})]$

That is, if the respondent is privately committed to either answer, we presume that they are disposed to communicate it. This presumption, as said, is context-sensitive. In other settings—e.g., a police interrogation or a televised interview—we may be far less inclined to assume that the respondent will disclose a commitment that could be incriminating or reputationally costly. In such contexts, partial opacity becomes socially intelligible. Importantly, these contextual shifts cannot be traced to a single conversational factor: topic sensitivity, interpersonal dynamics, institutional power asymmetries, and the perceived costs of disclosure all modulate the likelihood that Transparency holds. Even in informal contexts, certain questions may be experienced

²⁰ Specifically, in Krifka’s account, questions like (39) commit the speaker to the goal that the addressee will, without undue delay, commit to one of these propositions. This refinement does not affect our analysis.

as intrusive (e.g., regarding income, voting behaviour, medical status), thereby suppressing expectations of disclosure.

Sharing commitments is, in short, a socially regulated activity, governed not only by conversational norms but also by relational constraints. My claim here is that agents are sensitive to these constraints and deploy Transparency assumptions as regulative expectations to anticipate which private commitments are likely to be shared and which are not. When no overriding constraints are salient, Transparency may be presumed as a default means of increasing coordinative efficiency. It does so by narrowing the space of possible private commitments that the speaker might be withholding. This, in turn, has direct consequences for scalar reasoning. To see how, consider the following exchange:

- (41) BOSS: Were any of the jewels taken?
 EMPLOYEE: Some of them are missing.

Let us suppose that the following all count as proper answers to the boss's question:

- (42) a. None of them are missing. NONE
 b. Some of them are missing. SOME
 c. All of them are missing. ALL

In this context, it is plausible to assume that the employee would be disposed to share a private commitment to any of these propositions, if they held one. The Transparency assumptions in (43) are therefore reasonable to posit. Note that these assumptions do not entail that the employee holds a private commitment to any of these propositions (or to their negations). In particular, they remain fully compatible with verbal or non-verbal responses expressing uncertainty (e.g., "I don't know", shrugging).

- (43) $TRANS_{a,b}(\text{NONE})$, $TRANS_{a,b}(\text{SOME})$, $TRANS_{a,b}(\text{ALL})$

On our account, the employee's actual response is modelled as $C_{a,b}(\text{SOME})$ and Reflexivity licenses the inference $C_{a,a}(\text{SOME})$. From this, it follows that $C_{a,a}(\neg \text{NONE})$ and $C_{a,b}(\neg \text{NONE})$ both hold, which means that $TRANS_{a,b}(\text{NONE})$ is trivially met. But matters differ for $TRANS_{a,b}(\text{ALL})$ due to its interaction with the primary implicature associated with the employee's SOME-response. Recall that CB-Quantity yields:

- (44) $\neg Pt C_{a,b}(\text{ALL}) = \neg C_{a,a}(\text{ALL}) \vee \neg D_a(C_{a,b}(\text{ALL}))$

As discussed, this implicature does not, on its own, settle a 's private stance toward ALL. But consider now the Transparency assumption for ALL:

- (45) $TRANS_{a,b}(\text{ALL}) = [C_{a,a}(\text{ALL}) \rightarrow D_a(C_{a,b}(\text{ALL}))] \wedge [C_{a,a}(\neg \text{ALL}) \rightarrow D_a(C_{a,b}(\neg \text{ALL}))]$

This assumption rules out the possibility that $C_{a,a}(\text{ALL})$ holds if $\neg D_a(C_{a,b}(\text{ALL}))$ holds. As a result, combining (44) and (45) yields the inference $\neg C_{a,a}(\text{ALL})$. As before, this inference can be strengthened with the following committed speaker assumption:

$$(46) \quad CS_{a,a}(\text{ALL}) = C_{a,a}(\text{ALL}) \vee C_{a,a}(\neg \text{ALL})$$

Assuming that the speaker has a settled private stance on whether or not all the jewels are missing, we infer from $\neg C_{a,a}(\text{ALL})$ that $C_{a,a}(\neg \text{ALL})$ holds. By (45), this yields $D_a(C_{a,b}(\neg \text{ALL}))$. By the definition in (35), we conclude then that the speaker is in a position to communicate $\neg \text{ALL}$, that is, $PtC_{a,b}(\neg \text{ALL})$. The key inferential steps resulting from the interaction of primary implicature (PI), Transparency (Trans), and Committed Speaker (CS) assumptions can be summarised as follows:

$$(47) \quad \begin{aligned} \text{a. } & PI(\text{SOME}) \cap TRANS_{a,b}(\text{ALL}) \Rightarrow \neg C_{a,a}(\text{ALL}) \\ \text{b. } & CS_{a,a}(\text{ALL}) \cap TRANS_{a,b}(\text{ALL}) \Rightarrow D_a(C_{a,b}(\text{ALL})) \vee D_a(C_{a,b}(\neg \text{ALL})) \\ \text{c. } & PI(\text{SOME}) \cap TRANS_{a,b}(\text{ALL}) \cap CS_{a,a}(\text{ALL}) \Rightarrow C_{a,a}(\neg \text{ALL}) \wedge D_a(C_{a,b}(\neg \text{ALL})) \end{aligned}$$

Abstracting from this example, we can generalise the inference structure as follows:

1. In uttering S , a commits to b to act on the proposition p expressed by S : $C_{a,b}(p)$.
2. By Reflexivity, it follows from $C_{a,b}(p)$ that a is privately committed to p , hence $C_{a,a}(p)$.
3. By CB-Quantity, a would not have uttered S if they were in a position to communicate a stronger alternative proposition to p . Thus, for all $S' \in Alt(S)$, $\neg PtC_{a,b}(p')$.
4. If the Transparency assumption $Trans_{a,b}(p')$ holds, then $\neg C_{a,a}(p')$ follows.
5. If the Committed Speaker assumption $CS_{a,a}(p')$ also holds, then $C_{a,a}(\neg p')$ follows and, given $Trans_{a,b}(p')$, so does $PtC_{a,b}(\neg p')$.

This account preserves a key insight of our earlier attempt: Quantity implicatures do not directly reveal the speaker's private commitments. Rather, they track what the speaker is *not* in a position to commit to before the addressee. What distinguishes this account is the dual role attributed to Transparency assumptions in refining this initial step. First, when these assumptions are licensed, primary implicatures may be strengthened into conclusions about the speaker's private commitments. Second, once combined with CS assumptions, they may be used to interpret the negated alternative as content the speaker is in a position to commit to before the addressee. In the remainder of this section, I explore and illustrate this interaction in greater detail, extending the analysis to disjunctive statements.

4.4 Transparency and Committed Speaker

We turn now to three questions regarding the licensing, interaction and inferential implications of Transparency and CS assumptions. We begin with the issue of licensing. As discussed in Sauerland

(2004), the use of auxiliaries assumptions must be constrained to prevent contradictions and preserve interpretive coherence. Sauerland models this constraint via a consistency filter, whereby each auxiliary assumption is independently checked for consistency against the inference already drawn (cf. (10)). For our purposes, however, this constraint is too permissive. To see why, consider again (1), “I saw some animal in the woods”, and assume, for ease of exposition, the following scalar alternatives:

$$(48) \quad Alt(ANIMAL) = \left\{ \begin{array}{ll} \text{I saw a bear} & \text{BEAR} \\ \text{I saw a deer} & \text{DEER} \\ \text{I saw a coyote} & \text{COYOTE} \end{array} \right\}$$

Drawing the primary implicatures and introducing the Transparency assumption associated with these alternative yield the inferences $\neg C_{a,a}(\text{BEAR})$, $\neg C_{a,a}(\text{DEER})$, and $\neg C_{a,a}(\text{COYOTE})$. Consider now the potential CS assumptions: $CS_{a,a}(\text{BEAR})$, $CS_{a,a}(\text{DEER})$ and $CS_{a,a}(\text{COYOTE})$.²¹ Individually, each is consistent with what has been inferred so far. Taken together, however, they entail that the speaker is privately committed to $\neg \text{BEAR}$, $\neg \text{DEER}$ and $\neg \text{COYOTE}$, contradicting the Reflexive requirement that that the speaker is privately committed to ANIMAL . Moreover, arbitrarily choosing any proper subset of the CS assumptions would generate undesirable results. For instance, adopting $CS_{a,a}(\text{BEAR})$ and $CS_{a,a}(\text{DEER})$ (but leaving out $CS_{a,a}(\text{COYOTE})$) would yield $C_{a,a}(\neg \text{BEAR})$ and $C_{a,a}(\neg \text{DEER})$ and, therefore, the unwarranted conclusion that the speaker is privately committed to COYOTE .

These issues mirror the familiar problem of exclusion-of-alternatives in the grammatical approach to SIs (Fox 2007; Chierchia et al. 2012; Bar-Lev & Fox 2020; Marty & Romoli 2021, a.o.). A common solution in that tradition invokes Fox (2007)’s notion of *Innocent Exclusion*. Adapting this notion to the present framework yields our *Innocence* condition on auxiliary assumptions in (49). Under this condition, none of the CS assumptions in the example above are licensed if the corresponding Transparency assumptions have been introduced—as desired. We take this condition to be necessary both to maintain consistency and to block arbitrary strengthening.

(49) **INNOCENT ASSUMPTIONS** Let $\Sigma_{Alt(S)}$ be the maximal set of auxiliary assumptions of a given type associated with $Alt(S)$. An assumption $\sigma \in \Sigma_{Alt(S)}$ is licensed only if it is contained in the intersection of all maximal subsets of $\Sigma_{Alt(S)}$ that are consistent with:

- the common ground, and
- the inferences licensed by Quality and Quantity, and
- any innocent assumptions previously adopted.

²¹ The licensing issue is illustrated here by considering the introduction of CS assumptions in the course of scalar inferencing, but the issue is general and applies equally to Transparency assumptions.

This refinement brings us to our second question: how do Transparency and CS assumptions interact? In Section 4.3, we examined a case where both assumption types were simultaneously licensed. As the example above shows, however, the present theory also predicts that they may be mutually incompatible. We take this to be a virtue: it yields a fine-grained account of cases where scalar enrichments can proceed along different paths depending on which auxiliary assumptions are admissible. To illustrate, consider the following scenario:

- (50) **Voting intention:** A local election is approaching, with five candidates running for mayor. One week before the election, Billy asks his colleague Adam who he intends to vote for. Adam replies: “I will vote for Smith or Moore.”

By Reflexivity, Adam’s response licenses $C_{a,a}(\text{SMITH OR MOORE})$. By CB-Quantity, it follows that he is not in a position to communicate either of the following alternatives:

- (51) a. I will vote for Smith. SMITH
 b. I will vote for Moore. MOORE

From there, two inferential paths are available but incompatible: if we adopt the Transparency assumptions $\text{Trans}_{a,b}(\text{SMITH})$ and $\text{Trans}_{a,b}(\text{MOORE})$, then neither $\text{CS}_{a,a}(\text{SMITH})$ nor $\text{CS}_{a,a}(\text{MOORE})$ can be true; if we adopt the CS assumptions $\text{CS}_{a,a}(\text{SMITH})$ and $\text{CS}_{a,a}(\text{MOORE})$, then neither $\text{Trans}_{a,b}(\text{SMITH})$ nor $\text{Trans}_{a,b}(\text{MOORE})$ can be true. The first path yields an interpretation on which Adam would share his electoral choice if he had one, but he doesn’t know yet which of the two candidates he will vote for. The second yields an interpretation on which Adam knows which candidate he will vote for but refrains from disclosing his choice. These interpretations reflect two intuitive readings of Adam’s response: either he is genuinely undecided or he is strategically withholding a private commitment.

Finally, consider cases where Transparency and CS assumptions jointly support a transition from a primary implicature to a content-bearing conclusion, i.e., a shift from $\neg \text{PtC}_{a,b}(p')$ to $\text{PtC}_{a,b}(\neg p')$. On our account, $\text{PtC}_{a,b}(p)$ captures a readiness to publicly endorse a private commitment to p (cf. (35)). Such a disposition, while distinct from an explicit commitment, is not conversationally inert; it is itself a conversational move with norm-governing consequences. It shapes what the speaker may be held to, what the hearer may do next, and how common ground may develop. For instance, upon hearing that “Lupin stole some of the jewels”, Sholmes may treat his colleague as disposed to commit to $\neg \text{ALL}$ and respond accordingly, whether by promoting $\neg \text{ALL}$ into the common ground (52a), by requesting confirmation (52b), or by challenging the implicated proposition, (52c).

- (52) COLLEAGUE: Lupin stole some of the jewels.
 SHOLMES: Hmm, okay.

- a. I am curious as to why he didn't steal them all.
- b. So he didn't steal them all?
- c. But I doubt that he didn't steal them all—the ones you found must be decoys.

In sum, $PtC_{a,b}(p)$ captures communicative readiness—an inferable layer of stance beneath explicit commitments. Interlocutors may exploit this layer, for reasons of conversational economy or social strategy, to signal willingness (or reluctance) to endorse a proposition without doing so overtly. The interaction between Transparency and CS assumptions predicts when such readiness becomes publicly visible, and when it remains opaque.

4.5 Shades of opacity

Up to this point, we have examined cases in which Transparency and CS assumptions either work in tandem or block one another. The present account permits yet a more fine-grained view on which their interaction may vary across the alternatives activated by a single utterance. This flexibility, I argue, explains the otherwise puzzling interpretation of disjunctive statements in semi-opaque settings, such as Fox's Game Show scenario:

- (53) **Game Show (Fox 2014):** There are 100 boxes, and five of them contain a million dollars each (the rest are empty). The show's host knows the identity of the five boxes but, of course, is not permitted to disclose this information fully. At any point, contestants can take the risk of choosing a box. Throughout the game, hints are provided by the host, with the common understanding that these reveal only part of the relevant information available to the host. During one turn, the host said to the contestants: "There is money in box 20 or box 25". (20 OR 25)

Echoing previous remarks by Grice (1989), Fox (2014) notes that this disjunction is felicitous despite the host's evident knowledge of the prize locations, indicating that no speaker-ignorance inference arises in this context. Yet Fox observes that the usual exclusive inference, $\neg(20 \text{ AND } 25)$, remains available.²² As Fox discusses, this is unexpected under Neo-Gricean approaches, which treat ignorance and exclusivity inferences as derivationally linked: if NG-Quantity is operative, both inference types should arise (incorrectly predicting ignorance here); if suspended, neither should (failing to derive exclusivity here).

On the present account, by contrast, this pattern follows straightforwardly. Assuming the conversational maxims continue to apply, the primary implicatures are derived:

²² Grice (1989) noted that, in the context of a treasure hunt, a speaker known to know the prize's location can felicitously say, "The prize is either in the attic or in the garden." Fox's scenario highlights a new dimension: to the extent that *or* is interpreted exclusively here, this interpretation cannot be explained as a contextual entailment, unlike in the treasure hunt, where the uniqueness of the prize implies exclusivity. Experimental evidence supporting Fox's claim is provided by Agyemang (2020) and Marty et al. (2022).

- (54) a. $\neg PtC_{a,b}(20)$
 b. $\neg PtC_{a,b}(25)$
 c. $\neg PtC_{a,b}(20 \text{ AND } 25)$

Moreover, in (53), it is common ground that the host knows the box contents. Accordingly, the following CS assumptions already obtain:

- (55) a. $CS_{a,a}(20)$
 b. $CS_{a,a}(25)$
 c. $CS_{a,a}(20 \text{ OR } 25)$

As seen earlier, the CS assumptions about each disjunct blocks the corresponding Transparency assumptions. Here, in fact, the falsity of $TRANS_{a,b}(20)$ and $TRANS_{a,b}(25)$ follows independently of the primary implicatures in (54), simply from the host's utterance, the game rules and the given CS assumptions. Consider, for example, $TRANS_{a,b}(20)$:

- (56) $[C_{a,a}(20) \rightarrow D_a(C_{a,b}(20))] \wedge [C_{a,a}(\neg 20) \rightarrow D_a(C_{a,b}(\neg 20))]$

Since $CS_{a,a}(20)$ holds, the host must be committed either to 20 or to $\neg 20$. But communicating either proposition would reveal too much: endorsing 20 would reveal that box 20 wins outright; endorsing $\neg 20$ would indirectly reveal that box 25 wins. Either way, the required disposition for the host to publicly commit to 20 or to $\neg 20$ is impermissible. The same reasoning applies to the other disjunct. Therefore:

- (57) a. $\neg TRANS_{a,b}(20)$
 b. $\neg TRANS_{a,b}(25)$

Consider now the Transparency assumption associated with the conjunctive alternative:

- (58) $[C_{a,a}(20 \text{ AND } 25) \rightarrow D_a(C_{a,b} \text{ 20 AND 25})] \wedge [C_{a,a}\neg(20 \text{ AND } 25) \rightarrow D_a(C_{a,b} \neg(20 \text{ AND } 25))]$

This assumption is compatible with all prior inferences. For the same reasons as before, the host cannot be disposed to commit to 20 AND 25—communicating this proposition would immediately end the game. Thus, if (58) holds, $C_{a,a}(20 \text{ AND } 25)$ must be false and, given (55c), it follows that $C_{a,a}(\neg(20 \text{ AND } 25))$ must be true. Crucially, disclosing this private commitment to the participants is permissible: $\neg(20 \text{ AND } 25)$ rules out the possibility that both boxes contain money without identifying any specific box. Therefore, Transparency is selectively available for the conjunctive alternative:

- (59) $TRANS_{a,b}(20 \text{ AND } 25)$

Our account thus yields the following inferential pattern:

- (60) a. $\neg PtC_{a,b}(20) \wedge \neg PtC_{a,b}(\neg 20)$
 b. $\neg PtC_{a,b}(25) \wedge \neg PtC_{a,b}(\neg 25)$
 c. $PtC_{a,b}(\neg 20 \text{ AND } 25)$

In short, on this analysis, the host's utterance is semi-opaque: private commitments concerning the individual disjuncts remain hidden (no ignorance inference), while the commitment concerning their conjunction is transparently communicable (yielding exclusivity). This result aligns with the intuitions that, in contexts like (53), practical constraints shape, rather than undermine, cooperation: the host remains maximally informative within the norms of the game.

5 Discussion

Much of our ability to interpret and predict human behaviour rests on assumptions about what people are normatively expected and socially disposed to do. Revising and extending Geurts' commitment-based framework, I have argued that scalar inferences likewise may turn on such assumptions. The resulting model shifts the focus from reconstructing speakers' mental states and communicative intentions to tracking their private commitments and their dispositions to make these public. While preserving core insights from Gricean and Neo-Gricean accounts, this approach replaces a strongly mentalistic (or epistemic) model of scalar inferencing with a social-pragmatic interface structured by norms and dispositions. Conceptually, this shift reconfigures the explanatory architecture of scalar inferencing by changing what grounds SIs and how they relate to coordination and accountability in interaction. Empirically, this perspective provides a more fine-grained understanding of communicative affordances across utterance types, capturing gradations of opacity in guarded and strategically non-committal speech that have proven difficult for Neo-Gricean accounts. The proposal nevertheless remains programmatic in important respects. I therefore conclude by highlighting directions for further development.

Although the model is designed to apply across speech acts, the present discussion has focused primarily on assertions, with only brief remarks on imperatives and questions. Yet these other speech acts—most notably, questions—deserve closer scrutiny. A well-known inference associated with questioning is that the speaker (generally) lacks prior knowledge of the answer. Less discussed is that questions can also give rise to secondary SIs (though see Chemla 2008 and Romoli 2012). As shown in (61), the weak scalar term *some* in polar questions generally fails to trigger the usual exclusivity inference ($\neg \text{ALL}$), but its stronger scale-mate *all* may yield an inclusivity inference (*SOME*). This asymmetry mirrors the one observed in assertions where *some* and *all* appear under negation (e.g., “Not all of the jewels are missing” $\rightsquigarrow$ *some of the jewels are missing*). To my knowledge, the contrast in (61) remains a challenge for existing proposals.²³

²³ Romoli (2012) offers an interesting proposal for polar questions. Specifically, assuming a Hamblin-style semantics in which a polar question “*p?*” denotes the set $\{p, \neg p\}$ (Hamblin 1976), Romoli proposes that Quantity reasoning applies

- (61) a. Are some of the jewels missing? SOME?
 $\rightsquigarrow$ *the speaker doesn't know whether or not SOME*
 $\nrightarrow$ *the speaker assumes $\neg$ ALL*
- b. Are all of the jewels missing? ALL?
 $\rightsquigarrow$ *the speaker doesn't know whether or not ALL*
 $\rightsquigarrow$ *the speaker assumes SOME*

Let me add here a further observation: implicatures in polar questions appear to show the same contextual variability as those observed for disjunctive statements. Consider, for example, a teacher using an illustrated storybook about Detective Sholmes's stories and asking pre-schoolers, "Are all of the jewels missing?". The teacher is clearly not expressing ignorance but simply prompting children to commit either to ALL or to $\neg$ ALL. Yet the inclusivity inference (SOME) may still arise. Thus, as with disjunctive utterances in semi-opaque contexts, questions may yield secondary SIs even when ignorance inferences are absent. This suggests that Transparency assumptions can be selectively modulated across alternatives in questions, just as in disjunction. If so, the interplay between Transparency and Committed Speaker assumptions—central to the present account—may prove fruitful for questions as well. Much remains to be explored, but these initial observations suggest that a unified approach to SIs across speech acts may be desirable.

Another interesting direction, raised by one reviewer, is to explore how the present proposal might be imported into other architectures and, in particular, combined with grammatical approaches to SIs. For present purposes, I simply note that the proposal concerns primarily the nature of the attitude involved in scalar enrichment rather than the locus of its derivation. In particular, accounts of primary and ignorance implicatures based on covert epistemic operators—such as the Matrix K hypothesis of Meyer 2013—appear compatible with a commitment-theoretic reinterpretation in which the relevant operator expresses instead a social commitment indexed to the interlocutors (typically, speaker and addressee). Exploring a possible "Matrix C hypothesis", and the extent to which C might be understood as part of illocutionary structure, lies beyond the scope of this paper. Nevertheless, this possibility suggests that the conceptual shift advocated here is, to some extent, orthogonal to the question of whether scalar strengthening is derived through a conversational maxim or through covert exhaustification combined with auxiliary assumptions of the sort advocated here. Importantly, unlike Meyer's Matrix K axiom, which applies specifically to *assertively used sentences*, a commitment-theoretic analogue would naturally extend beyond assertion to other illocutionary types for which epistemic attitudes are not the appropriate objects (e.g., directives and, arguably, questions).

to both elements of this set. On his account, a question "p?" implicates *q* if both "p" and " $\neg$ p" would entail or implicate *q*. Thus, this account can explain the implicature in (61b). However, by the same reasoning, it also predicts that (61a) should implicate $\neg$ ALL, since both SOME and $\neg$ SOME would equally implicate or entail $\neg$ ALL. This prediction does not seem to be borne out.

A third line of extension concerns cross-cultural variability. The present account assumes that successful communication hinges on agents' ability to coordinate action through the alignment of private and public commitments. On this view, the Reflexivity of commitment and the commitment-based maxim of Quantity serve to minimise misalignment and facilitate coordination. Since efficient coordination is arguably a universal need of human societies, one might expect these principles to operate across cultures. At first glance, however, some ethnographic data seem to challenge this expectation. Keenan (1976), investigating conversational practices in Malagasy society, famously observes that Malagasy interlocutors routinely provide less information than their conversational partner requests, even when they possess the relevant information. Her canonical example is illustrative:

If A asks B, "Where is your mother?" and B responds, "She is either in the house or at the market", B's utterance is not usually taken to imply that B is unable to provide more specific information needed by the hearer. The implicature is not made, because the expectation that the speakers will satisfy informational needs is not a basic norm. (Keenan 1976)

Contrary to the expectation that speakers should be maximally informative, Malagasy interlocutors often withhold information, especially when the information is significant (costly to obtain or to disclose) or when speaker and hearer are not socially close. According to Keenan, this conversational practice reflects a cultural norm in which the possession of significant information confers social prestige, and its indiscriminate disclosure may jeopardise social capital or face.

While Keenan took these findings to suggest culturally variable adherence to the Gricean maxims, the present model offers a more conservative alternative. It allows us to maintain that the same commitment-based principles govern cooperative communication across cultures, while explaining the observed variation in terms of differences in Transparency assumptions: expectations, licensed by social norms, about which private commitments are liable to be shared, and in which contexts. As Keenan herself notes, factors modulating scalar reasoning in Malagasy society—such as the significance of the information, its face-threatening potential, and the nature of the speaker–hearer relationship—also modulate scalar reasoning in Western societies, albeit less dramatically. These are precisely the kind of factors that, on our account, affect whether Transparency is assumed. Thus, rather than posing culture-specific adjustments to the maxims themselves, the present framework can preserve the generality of Quantity while accommodating variation in the socially grounded norms that regulate commitment sharing. In this way, it may strike a balance between the universal and the particular—between the normative fabric of communication and the social contexts in which it is woven.

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